

FLIPKART – DATA ARCHITECTURE & DAX STRATEGY DOCUMENT

E-Commerce Customer Orders & Product Sales Analytics | Power BI

Purpose: provide a handover-ready architecture reference covering source, tables, facts/dimensions, columns, data types, relationships, schema, ER view, Power BI connectivity, references, and a DAX implementation strategy

Document Item	Value
Project Title	Flipkart Analytics
Platform	Microsoft Power BI
PBIX reviewed	Flipkart.pbix
Report pages	Executive Dashboard, Tooltip & Sales Vs Profit
Primary business scope	Orders, sales, profit, customers, products, delivery and payment analysis
Document purpose	Project handover, interview explanation, portfolio reference and future maintenance

Validation principle: this document distinguishes what is visible in the supplied PBIX definition from items that should be validated directly in Power BI Desktop/Data view.

1. Executive Summary

The Flipkart Power BI project is an e-commerce analytics dashboard focused on customer orders, product performance, sales, profit, returns, delivery status, payment mode and geographic analysis. The supplied PBIX contains a main Executive Dashboard page, Tooltip page and Sales Vs Profit drill through page.

- The main Executive Dashboard is the primary management/analytics view and contains KPI cards, charts, slicers, navigation buttons and supporting visuals.
- The Sales Vs Profit page is configured as a Drillthrough page using Dim Product[Brand] as the drillthrough field.
- The Tooltip page is hidden in view mode and contains cards plus a table for contextual hover information.
- The model uses four core semantic tables visible in the PBIX diagram: Fact Orders, Dim Customer, Dim Product and Dim Date, plus a Metric Selection helper table.
- The project uses bookmark-driven navigation/visual-state interactions and a Metric Selection measure for dynamic metric analysis on the Sales Vs Profit page.

2. Source & Evidence

Item	Observed in PBIX	Status
Report / dashboard heading	E-Commerce Customer Orders & Product Sales Analytics Dashboard	Confirmed
Core fact table	Fact Orders	Confirmed
Dimension tables	Dim Customer, Dim Product, Dim Date	Confirmed
Helper table	Metric Selection	Confirmed
Report pages	Executive Dashboard; Tooltip and Sales Vs Profit	Confirmed
Bookmarks	10 bookmark definitions	Confirmed
Drillthrough	Sales Vs Profit page bound to Dim Product[Brand]	Confirmed
Tooltip	Tooltip page with cards and a table	Confirmed
Original physical source	Not exposed by the supplied PBIX report-definition extraction	Needs validation
Exact model relationship cardinality	Not reliably exposed in the extracted layout	Needs validation
Exact DAX expression text	Measure references are visible; full expression text is not exposed in the extracted report definition	Needs validation
RLS rules	No explicit RLS configuration could be confirmed from the report-definition	Needs validation

	artifacts reviewed	
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3. Business Scope

Area	Questions the report supports
Customers	How many customers are represented and how are they distributed geographically?
Orders	How are order volumes distributed across time, location and payment mode?
Sales	How much sales value is generated and how does it vary by product, brand, state, zone and time?
Profit	Which states/zones/brands/products contribute to profit?
Products	Which categories, sub-categories, products and brands drive sales/profit?
Delivery	How are delivery status and shipping-related fields represented in the detailed analysis?
Payments	How do payment modes contribute to order volume?

4. Data Architecture

The extracted PBIX diagram contains Fact Orders, Dim Customer, Dim Product, Dim Date and Metric Selection. The recommended semantic pattern for this project is a star-schema style model, with dimensions filtering the transaction fact.

Source → Power Query → Dimensional Model → DAX Measures → Power BI Report → Power BI Service

Table	Role	Business purpose
Fact Orders	FACT	Order-level transactional values such as sales, profit, quantity, order date, payment and delivery fields.
Dim Customer	DIMENSION	Customer attributes and geographic/customer segmentation fields.
Dim Product	DIMENSION	Product, category, sub-category, brand, MRP and rating attributes.
Dim Date	DIMENSION	Year, month, month name and day attributes for time analysis.
Metric Selection	HELPER / DISCONNECTED	Supports dynamic metric selection on the Sales Vs Profit page.

4.1 Conceptual Star Schema

Dimension / Helper	Key / main fields	Fact relationship concept
Dim Date	Date-related fields	Filters Fact Orders by order date
Dim Customer	Customer ID	Filters Fact Orders by Customer ID
Dim Product	Product ID	Filters Fact Orders by Product ID
Fact Orders	Order ID, Product ID, Customer ID, Order Date	Central transaction table
Metric Selection	Metric, Selected Metric	Disconnected helper used for dynamic selection

The relationship direction/cardinality should be verified in Power BI Model view before using this document as a formal database specification.

5. Column Inventory

Dim Date

Column	Data type	Typical usage
Day	Validate in Data view	Time filtering and trends
Month	Validate in Data view	Time filtering and trends
Year	Validate in Data view	Time filtering and trends

Dim Customer

Column	Data type	Typical usage
Country	Validate in Data view	Customer / geography analysis
State	Validate in Data view	Customer / geography analysis
City	Validate in Data view	Customer / geography analysis
Zone	Validate in Data view	Customer / geography analysis
Pin code	Validate in Data view	Customer / geography analysis
Customer ID	Validate in Data view	Customer / geography analysis
Customer Name	Validate in Data view	Customer / geography analysis
Gender	Validate in Data view	Customer / geography analysis
Phone	Validate in Data view	Customer / geography analysis
Email	Validate in Data view	Customer / geography analysis
Membership Type	Validate in Data view	Customer / geography analysis
Total Customers	Validate in Data view	Customer / geography analysis

Dim Product

Column	Data type	Typical usage
Product ID	Validate in Data view	Product / category analysis
Product Name	Validate in Data view	Product / category analysis
Brand	Validate in Data view	Product / category analysis
Category	Validate in Data view	Product / category analysis
Sub-Category	Validate in Data view	Product / category analysis
MRP (INR)	Validate in Data view	Product / category analysis
Rating	Validate in Data view	Product / category analysis

Fact Orders

Column	Data type	Typical usage
Delivery Status	Validate in Data view	Transaction / KPI / filtering
Discount (%)	Validate in Data view	Transaction / KPI / filtering
Order Date	Validate in Data view	Transaction / KPI / filtering
Order ID	Validate in Data view	Transaction / KPI / filtering
Payment Mode	Validate in Data view	Transaction / KPI / filtering
Product ID	Validate in Data view	Transaction / KPI / filtering
Profit (INR)	Validate in Data view	Transaction / KPI / filtering
Profit Margin %	Validate in Data view	Transaction / KPI / filtering
Quantity	Validate in Data view	Transaction / KPI / filtering
Sales (INR)	Validate in Data view	Transaction / KPI / filtering
Ship Date	Validate in Data view	Transaction / KPI / filtering
Ship Mode	Validate in Data view	Transaction / KPI / filtering
Total Orders	Validate in Data view	Transaction / KPI / filtering
Total Returns	Validate in Data view	Transaction / KPI / filtering
Unit Price (INR)	Validate in Data view	Transaction / KPI / filtering

Metric Selection

Column	Data type	Typical usage
Metric	Validate in Data view	Dynamic metric selection

Selected Metric	Validate in Data view	Dynamic metric selection
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6. KPI & DAX Strategy

The PBIX uses both direct aggregations and model measures. The following section documents the measure names that are visibly referenced by the report and provides implementation patterns. Where the exact stored DAX expression is not exposed by the supplied extraction, the formula is labelled as a recommended/validation pattern rather than claimed as the original expression.

Measure / KPI	Role	Recommended DAX pattern
Total Orders	Order volume	SUM('Fact Orders'[Total Orders]) or DISTINCTCOUNT('Fact Orders'[Order ID]) — validate intended business definition
Total Sales	Sales value	SUM('Fact Orders'[Sales (INR)])
Total Profit	Profit value	SUM('Fact Orders'[Profit (INR)])
Profit Margin %	Profitability ratio	DIVIDE([Total Profit], [Total Sales])
Total Returns	Return KPI	Validate stored measure logic; report references Fact Orders[Total Returns] as a measure
Total Customers	Customer count	DISTINCTCOUNT('Dim Customer'[Customer ID]) or the stored measure
Selected Metric	Dynamic metric	SWITCH/SELECTEDVALUE pattern driven by 'Metric Selection'[Metric]

Total Sales =
SUM('Fact Orders'[Sales (INR)])

Total Profit =
SUM('Fact Orders'[Profit (INR)])

Profit Margin % =
DIVIDE([Total Profit], [Total Sales])

Total Customers =
DISTINCTCOUNT('Dim Customer'[Customer ID])

6.1 Dynamic Metric Selection

The Sales Vs Profit page contains a Metric Selection slicer and a Selected Metric measure. The selected measure is used as the value axis of a clustered bar chart by Brand.

Selected Metric =
SWITCH(
 SELECTEDVALUE('Metric Selection'[Metric]),
 "Sales", [Total Sales],
 "Profit", [Total Profit],
 "Orders", [Total Orders],
 [Total Sales]
)

The exact option labels stored in the Metric Selection table should be checked in Data view. The pattern above is an implementation template, not a claim about the exact original DAX.

7. Report / Visual Reference

Page	Visual inventory	Total visuals
Executive Dashboard	slicer (18), actionButton (12),	73

	cardVisual (12), image (9), donutChart (4), lineStackedColumnComboChart (3), clusteredColumnChart (2), pieChart (2), textbox (2), clusteredBarChart (2), areaChart (1), barChart (1), lineClusteredColumnComboChart (1), tableEx (1), shape (1), funnel (1), lineChart (1)	
Tooltip	cardVisual (3), tableEx (1)	4
Sales Vs Profit	clusteredBarChart (2), slicer (2), actionBar (1), tableEx (1)	6

7.1 Executive Dashboard

The Dashboard is the main report page. The PBIX contains 73 visual objects on this page, including KPI cards, slicers, charts, navigation buttons, images, a table, a funnel and supporting shapes/text.

Visual / analysis	Fields observed in PBIX
Profit by State	Dim Customer[State] + Fact Orders[Profit (INR)]
Orders by Year	Dim Date[Year] + Fact Orders[Total Orders]
Orders by Payment Mode	Fact Orders[Payment Mode] + Fact Orders[Total Orders]
Sales by Day	Dim Date[Day] + Fact Orders[Sales (INR)]
Profit by Zone	Dim Customer[Zone] + Fact Orders[Profit (INR)]
Orders by City	Dim Customer[City] + Fact Orders[Total Orders]
Sales by Product	Dim Product[Product Name] + Fact Orders[Sales (INR)]
Sales vs Profit by Brand	Dim Product[Brand] + Fact Orders[Sales (INR)] + Fact Orders[Profit (INR)]
Customer/order comparison by State	Dim Customer[State] + Fact Orders[Total Orders] + Dim Customer[Total Customers]
Category/Product filters	Dim Product[Category], Dim Product[Product Name]
Zone filters	Dim Customer[Zone]

7.2 Sales Vs Profit

This page is configured as a Drillthrough page. Its drillthrough filter is Dim Product[Brand]. It contains a dynamic metric chart, a Sales vs Profit comparison chart, a detail table, a Sub-Category slicer and a Metric slicer.

Visual	Fields / function
Dynamic clustered bar chart	Dim Product[Brand] + Metric Selection[Selected Metric]
Sales & Profit by Brand	Dim Product[Brand] + Fact Orders[Sales (INR)] + Fact Orders[Profit (INR)]
Detail table	State, City, Zone, Category, Product Name, Brand, Sales, Profit, Profit Margin
Metric slicer	Metric Selection[Metric]
Sub-Category slicer	Dim Product[Sub-Category]
Drillthrough field	Dim Product[Brand]

7.3 Tooltip

The Tooltip page view mode and has a 700 × 385 design canvas. It contains three cards and one table, using Product Name, Sales (INR) and Profit (INR) for contextual detail.

8. Bookmark & Navigation Design

Bookmark	Purpose / observed state
Dashboard Bookmark	Dashboard navigation / visual-state interaction
Order Bookmark	Dashboard visual-state interaction
Order Home	Dashboard navigation / visual-state interaction
Profit Bookmark	Dashboard visual-state interaction
Profit Home	Dashboard navigation / visual-state interaction
Sales Bookmark	Dashboard visual-state interaction
Sales Home	Dashboard navigation / visual-state interaction

Sales vs Profit Bookmark	Dashboard visual-state interaction
Total Bookmark	Dashboard visual-state interaction
Total Home	Dashboard navigation / visual-state interaction

The PBIX contains ten bookmark definitions: Order Bookmark, Order Home, Sales Bookmark, Sales Home, Profit Bookmark, Profit Home, Total Bookmark, Total Home, Sales vs Profit Bookmark and Dashboard Bookmark. Several bookmarks control visual display states and filters on the Dashboard.

9. Interactivity & UX Features

- Bookmark-based navigation and visual-state switching on the Dashboard.
- Drillthrough from the main report to the Sales Vs Profit page using Brand.
- Hidden Tooltip page for contextual hover detail.
- Metric Selection slicer + Selected Metric measure for dynamic metric switching.
- Slicers for date, product category/product name, zone and sub-category.
- Action buttons for navigation and home/back-style interactions.
- The report definition uses enhanced tooltips and allows filter-type changes.

10. Power BI Build / Connectivity Workflow

1. Open Power BI Desktop and load the actual source used by the project.
2. Use Power Query to standardize column names, data types, null values and dates.
3. Create/validate Dim Date, Dim Customer and Dim Product dimensions.
4. Load Fact Orders at the correct business grain.
5. Create/validate relationships in Model view.
6. Create reusable base measures before ratio/time-intelligence measures.
7. Build the Dashboard visuals and slicers.
8. Configure bookmarks, buttons and navigation.
9. Configure the Sales Vs Profit drillthrough page.
10. Configure the hidden Tooltip page.
11. Validate totals against the source before publishing.

11. Validation Checklist

Check	Status / action
Source connection	Validate actual Excel/CSV/SQL/cloud source in Power Query.
Fact Orders grain	Validate exact row grain and business key in Data/Model view.
Data types	Validate dates, numeric fields, IDs and percentage fields.
Relationships	Validate key pairs, cardinality and active relationships.
Total Orders definition	Validate whether the stored field represents row-level 1s, order counts or another business definition.
Total Returns measure	Open measure and confirm its stored DAX.
Profit Margin % measure	Open measure and confirm its stored DAX.
Total Customers measure	Open measure and confirm its stored DAX.
Metric Selection options	Check the exact values stored in the helper table.
Drillthrough	Test Brand drillthrough from source visual to Sales Vs Profit.
Tooltip	Test tooltip rendering and filter context.
Bookmarks	Test all 10 bookmarks after any visual/model change.
Privacy	Review Table page because customer contact fields are displayed.
RLS	Validate whether row-level security is required and configured in the published semantic model.
Refresh	Run manual refresh successfully before scheduled refresh.

12. Power BI Service / Deployment Checklist

- Publish the PBIX to the intended Power BI workspace.
- Confirm the semantic model refresh succeeds.
- Configure gateway/credentials if the real source requires them.
- Run Refresh now successfully before enabling a schedule.
- Validate report filters, bookmarks, drillthrough and tooltip behavior in the Service.
- Validate access to the detailed Table page because it contains customer contact information.
- Configure RLS only after the business security rules are defined and tested.
- Publish through an App when controlled business-user navigation is required.

13. Project Explanation for Interview / Resume

Built an interactive Power BI e-commerce analytics dashboard for customer orders and product sales analysis. The project uses a dimensional semantic model with Fact Orders, Dim Customer, Dim Product and Dim Date, plus a Metric Selection helper for dynamic metric switching. The report includes KPI cards, slicers, geographic and product analysis, bookmark-driven navigation, a custom tooltip page, a Brand-based drillthrough page, dynamic metric selection and a detailed order table.

- Power Query: data cleaning preparation and standardization.
- Data Modeling: fact/dimension structure and relationship validation.
- DAX: reusable KPI measures, ratios and dynamic metric selection.
- Visualization: KPI cards, bar/column charts, combo charts, pie/donut charts, slicers and detail tables.
- Interactivity: bookmarks, buttons, drillthrough and tooltip.
- Deployment: Power BI Service refresh and access validation.

14. Final Technical Summary

Layer	Flipkart project implementation
Source	Actual physical source not exposed by the supplied PBIX extraction; validate in Power Query.
Transformation	Power Query layer should standardize dates, nulls, types and keys.
Model	Fact Orders + Dim Customer + Dim Product + Dim Date + Metric Selection helper.
Semantic / DAX	Sales, Profit, Profit Margin, Orders, Customers, Returns and dynamic Selected Metric logic.
Report	Dashboard + Tooltip + Sales Vs Profit drill through
UX	Bookmarks, action buttons, slicers and dynamic metric selection.
Service	Publish, refresh, security/access validation and report testing.